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16 May 2026

University of Maryland Global Campus
Adjunct Faculty Hiring — Artificial Intelligence
Grafenwöhr, Germany

Re: Adjunct Faculty Position — Artificial Intelligence

Dear Hiring Committee,

I am applying for the Adjunct Faculty position in Artificial Intelligence. My subject-matter depth spans the full stack of what an AI curriculum covers — from mathematical foundations and model training to agentic system design — and I have built original systems at each layer, including a novel operating system with its own theoretical framework and kernel language.

At the systems level, Buhera is a research OS whose architecture rests on three theorems I developed: the Triple Equivalence (observation, computation, and processing are formally identical operations), the S-Entropy Uniqueness Theorem (a provably unique continuous embedding for backward trajectory navigation), and the Circularity Theorem (why content-based self-preparation cannot produce a non-circular progress measure). The kernel includes a Proof Validation Engine using Lean 4, a Penultimate State Scheduler that prioritises processes by distance to goal state rather than by time slice, and a declarative kernel language (vaHera) in which programs specify final states rather than instruction sequences. Validated end-to-end: 100% accuracy on 32 NIST compounds, $O(\log_3 N)$ retrieval latency, $122,640\times$ speedup over discrete search at the embedding boundary. Six peer-review-ready papers are in the repository. This gives me the ability to teach AI foundations — including computability, complexity, and the theoretical limits of what learning systems can and cannot do — from first-hand derivation rather than textbook summary.

At the model training level, Purpose formalises the **Backward Training Principle** — a formal inversion of conventional easy-to-hard curriculum learning proved strictly optimal for any domain admitting a *pure-intent regime*. The central result: a policy ϵ -competent on the pure-intent regime (scalar objective, full capability envelope exercised, unambiguous feedback) is ϵ -competent on every constrained sub-regime by *feasibility inclusion*, not by generalisation. A Forward Training Collapse Theorem establishes a provably positive Vapnik–Chervonenkis failure gap for the conventional direction; a Sample Complexity Theorem gives $(N + 1)\times$ sample savings for a cascade of depth N , and $2^{N-1}\times$ for binary-nested cascades. The origin was concrete: I derived the theoretical minimum lap time for a Formula circuit and realised the circuit is the pure-intent regime for all driving sub-tasks — road, urban, and parking driving are strict feasibility restrictions of it. The principle generalises to protein design, scientific query compilation, and any domain with additive constraints and a natural extremal objective. This is a result I can teach from

first derivation: not “start easy,” but “start at the fully-resolved objective and project downward.”

At the interface and agentic layer, Zangalewa is the **Minimum Sufficient Interceptor** (MSI) sitting at the boundary of the Buhera OS. The OS presents a blank screen — no windows, no running applications. A user types; the MSI intercepts the utterance, extracts semantic coordinates in a bounded $[0, 1]^3$ partition space via a hierarchical k -ary specialist cascade requiring $O(\log_k N)$ small-model invocations, and hands typed coordinates to the vaHera kernel. A **Sufficiency Theorem** proves the interceptor’s parameter count is $O(D \cdot b \cdot \log |\Sigma|)$ — critically independent of the downstream substrate’s state-space size: an OS-scale substrate needs $\sim 10^7$ – 10^8 parameters, not a 100 B-parameter model. Session trajectory maintenance via precision-by-difference timing and a proved-complete four-state focus-arbitration machine (**Blank/Prompt/Artifact/Artifact + Prompt**) complete the mediation architecture. The autonomous code-repair pipeline that AI students typically first encounter — exception capture $\rightarrow$ LLM diagnosis $\rightarrow$ patch $\rightarrow$ git-isolated apply $\rightarrow$ rollback — is a secondary domain application of this same architecture. Four-Sided-Triangle is a separate 8-stage RAG system demonstrating hierarchical cascade routing at the document-retrieval level, with a Rust backend providing 10–50 $\times$ speedup.

UMGC’s student population — military personnel and working professionals — benefit most from instruction that connects abstract concepts to real engineering decisions. I can teach why attention is $O(n^2)$ in sequence length and what that means for practical deployment. I can explain why RAG exists as a solution to a specific failure mode in fine-tuning, and when you should fine-tune instead. I can walk through why autonomous agents fail in ways that simple chatbots do not, using failure modes I have encountered in systems I built. This is a different kind of instruction from someone who has studied these topics; it is instruction from someone who has made the mistakes.

I hold a Master’s in Computational Biology (Universität Tübingen) and completed doctoral research in Computational Lipidomics at TUM. I am legally resident in Germany with full work authorisation and am available immediately. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye